

SDR Hunter — Quick Start

1. Install

```
# Python deps (works without SDR hardware using the built-in mock device)
pip install -r requirements.txt

# Full SDR driver stack (SoapySDR + BladeRF/RTL-SDR/HackRF/LimeSDR/Pluto/USRP/SDRplay/
Airspy/NooElec)
./install_drivers.sh
```

2. Run

```
./launch.sh                # desktop GUI (same as: python3 main.py --gui)
python3 main.py --web       # web interface only (http://localhost:8000)
python3 main.py --both      # GUI + web server
python3 main.py --list-devices
```

No SDR attached? The app automatically falls back to a synthetic **mock device**, or force it with:

```
SDRHUNTER_FORCE MOCK=1 ./launch.sh
```

3. First scan

1. Pick a device in the **Device** toolbar drop-down.
2. Set **RX0** center frequency (e.g. 100.0 MHz) and sample rate.
3. Press ► **Start Scan** (or Space).
4. Detected signals appear in the right-hand list; double-click one to focus **RX1**.

4. Tabs

- **Main** — dual-RX spectrum + waterfall + signal list
- **Drone Tracking** — map + Remote ID / OpenDroneID contacts (F3)
- **Audio Decoder** — AM/FM/SSB demod + classification
- **Weather Satellite** — NOAA APT / METEOR-M2 LRPT
- **Spectrum Hunting** — band-hopping hunt mode (F2)
- **Signal Database** — browse/search 90+ known signals

5. Keyboard shortcuts

Key	Action
F1	Standard view
F2	Spectrum Hunting view
F3	Drone Tracking view
F4	Dual Signal Analysis
Space	Start/Stop scanning
R	Start/Stop recording
Ctrl+T	Tune RX1
Ctrl+B	Bookmark current frequency
Ctrl+L	Load baseline
Esc	Cancel current operation

6. Remote / web access

Enable the  **Web Server** toolbar button (or `--web`) and browse to `http://<host>:8000` from another machine on the network.